

EDTF: A Plug-and-Play Learning Framework for Estimated Delivery Time Task

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Abstract

Accurate estimation of delivery time (EDT) is a critical factor in web e-commerce user experience. The pursuit of higher EDT accuracy has predominantly centered on designing increasingly complex model architectures. While valuable, this architecture-centric paradigm creates a tension between its high iteration costs and the industrial demand for agile deployment. This work, therefore, explores a complementary dimension: enhancing model performance by optimizing the learning process itself. We propose EDTF, a novel, plug-and-play composite learning framework that empowers existing models by augmenting their learning objective. EDTF first transforms the traditional regression problem into a structured ordinal classification task to address the training difficulties inherent in direct regression and preserve temporal order. It then introduces a cross-view consistency paradigm, decomposing the prediction task into two related views: the macroscopic end-to-end delivery time and the microscopic next-hop duration. By enforcing a self-supervised signal that aligns the sum of future next-hop durations with the overall EDT, our framework enables models to learn more robust temporal representations without extra features. Extensive experiments on a large-scale industrial dataset show that EDTF, as a plugin, consistently enhances performance and accelerates convergence across five diverse architectures. Critically, an EDTF-optimized model has been successfully deployed in a live production environment, demonstrating significant improvements over its predecessor. This work thus presents a validated and valuable new paradigm for the economical and efficient application of web services reliant on trajectory-based forecasting, from e-commerce to ride-hailing and food delivery.

CCS Concepts

• Applied computing → Online shopping; • Information systems → Data mining.

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Keywords

Estimated Delivery Time, Plug-and-Play, Composite Learning Framework, Web E-commerce

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1 Introduction

In modern e-commerce, an accurate Estimated Delivery Time (EDT) is a critical factor that directly influences consumer purchasing decisions, enhances user experience, and generates significant economic value [3, 9]. Consequently, the pursuit of higher accuracy has spurred a major and fruitful line of research focused on designing increasingly complex model architectures, from early spatiotemporal networks like DeepTTE [6], to attention-based models such as FDNET [2] and MTSTAN [10], and sophisticated graph-based architectures including M²G4RTP [1] and I2RTP [4].

However, this architecture-centric approach faces two fundamental challenges. First, its performance gains eventually plateau, as they are ultimately capped by the expressive power of a given architecture and its features. Our work provides strong empirical evidence that this performance ceiling is not absolute and can be significantly raised by optimizing the learning process itself, without altering the model or introducing new features. Second, beyond this performance limitation, the practical challenges of this approach become apparent in fast-paced, web-scale industrial deployments. Pursuing performance gains by adopting a new architecture is a heavyweight operation, often incurring substantial costs from re-engineering data pipelines and implementing new models from scratch. This conflict between the high cost of architectural innovation and the need for agile deployment has become a critical bottleneck. In response, we argue that the focus of innovation should shift from designing more complex architectures to shaping more efficient learning processes. Adhering to this principle, we propose a novel, plug-and-play composite learning framework, termed EDTF (Efficient Estimated Delivery Time Framework). The core idea of EDTF is to reformulate the learning objective itself, thereby empowering any existing model to automatically learn

more robust and generalizable representations. This approach enables systematic improvements in prediction accuracy, convergence speed, and robustness at a minimal marginal cost, without altering the underlying model architecture or feature engineering. Our primary contributions are:

1. The design of EDTF, a novel, plug-and-play learning framework tailored for bridging the gap between research and large-scale web applications.
2. Extensive experiments on a massive, real-world dataset from the real online shop platform, demonstrating that EDTF consistently yields significant performance gains when applied to diverse deep learning models.
3. The successful deployment of an EDTF-enhanced model in a live production environment, which achieved a huge improvement over the previous version, validating its real-world value and feasibility.

2 Methodology

2.1 Problem Formulation

A package's logistics journey is represented by a temporally ordered sequence of events $\mathcal{U} = (u_1, u_2, \dots, u_T)$, where the final event u_T signifies the successful delivery as shown in Fig 1. A key characteristic of such real-world web data is that the time intervals between consecutive events are irregular and highly variable. For each event u_t in the sequence (where $1 \leq t < T$), our goal is to predict the package's future trajectory from two distinct but intrinsically related perspectives, which we term the **microscopic** and **macroscopic** views:

1. The Microscopic View (Next-Hop Duration): This perspective focuses on the immediate next step. We aim to predict the time from the current event u_t to the very next event u_{t+1} . The ground-truth for this is the actual duration, denoted as $\Delta h_t \in \mathbb{R}^+$.

2. The Macroscopic View (End-to-End Delivery Time): This perspective takes a holistic, long-term view of the entire remaining journey. The ground-truth is the total actual time from the current event u_t until the final delivery event u_T , denoted as $\Delta d_t \in \mathbb{R}^+$. Our model's prediction for this value is referred to as the **Estimated Delivery Time (EDT)**.

The objective is to train a model f_θ that takes a feature vector $\mathbf{x}_t$ corresponding to event u_t and produces accurate predictions for both the next-hop duration and the EDT. We treat EDT as the primary task and the next-hop duration prediction as an auxiliary one. To achieve this, we reformulate the problem as classification. The continuous ground-truth durations Δh_t and Δd_t are discretized into N_h and N_d categories, yielding labels $y_{h,t} \in \{1, \dots, N_h\}$ and $y_{d,t} \in \{1, \dots, N_d\}$. The model f_θ then outputs two logits vectors at each time step t : $\mathbf{z}_{h,t} \in \mathbb{R}^{N_h}$ for the next-hop duration and $\mathbf{z}_{d,t} \in \mathbb{R}^{N_d}$ for the EDT, which represent the model's estimates in a probabilistic format.

2.2 The EDTF Learning Framework

The EDTF framework guides the optimization of f_θ via a composite loss function, $\mathcal{L}_{\text{EDTF}}$. This function comprises three core components: a base classification loss $\mathcal{L}_{\text{cls}}$, a consistency loss $\mathcal{L}_{\text{cons}}$ to enforce a cross-view constraint, and an auxiliary regression loss

$\mathcal{L}_{\text{reg}}$ to supervise the predicted expectation.

$$\mathcal{L}_{\text{EDTF}} = \mathcal{L}_{\text{cls}} + \lambda_{\text{cons}} \mathcal{L}_{\text{cons}} + \lambda_{\text{reg}} \mathcal{L}_{\text{reg}} \quad (1)$$

where λ_{cons} and λ_{reg} are hyperparameters that balance the loss terms. Each component is detailed below.

2.2.1 Structured Classification. To mitigate the outlier sensitivity of direct regression and produce richer probabilistic outputs, we first reformulate the prediction task as ordinal classification. The base classification loss, $\mathcal{L}_{\text{cls}}$, is the standard cross-entropy applied to both the next-hop duration and EDT predictions:

$$\mathcal{L}_{\text{cls}} = \frac{1}{|\mathcal{M}|} \sum_{t \in \mathcal{M}} (\mathcal{L}_{\text{CE}}(\mathbf{z}_{h,t}, y_{h,t}) + w_{d,t} \cdot \mathcal{L}_{\text{CE}}(\mathbf{z}_{d,t}, y_{d,t})) \quad (2)$$

where $\mathcal{L}_{\text{CE}}(\mathbf{z}, y) = -\log \left(\frac{\exp(\mathbf{z}[y])}{\sum_{j=1}^N \exp(\mathbf{z}[j])} \right)$, $\mathcal{M}$ is the set of valid time steps and $w_{d,t}$ is a weight at time step t . To incorporate the inherent ordinal relationship between time categories, we then derive a continuous prediction by computing the probabilistic expectation of the class indices:

$$\hat{y}(z) = \sum_{i=1}^N i \cdot \text{Softmax}(\beta \cdot z_i) \quad (3)$$

where β is an inverse temperature. This yields continuous predictions for the next-hop duration ($\hat{h}_t$) and EDT ($\hat{d}_t$). We then introduce an auxiliary regression loss, $\mathcal{L}_{\text{reg}}$, to directly supervise these expected values, using a log transformation for stability:

$$\mathcal{L}_{\text{reg}} = \frac{1}{|\mathcal{M}|} \sum_{t \in \mathcal{M}} \left(\|\log(\hat{h}_t + 1) - \log(\Delta h_t + 1)\|_2^2 + \|\log(\hat{d}_t + 1) - \log(\Delta d_t + 1)\|_2^2 \right) \quad (4)$$

2.2.2 Cross-View Consistency as a Self-Supervised Signal. The core innovation is a self-supervised signal derived from the intrinsic relationship between the microscopic (next-hop duration) and macroscopic (EDT) views. Intuitively, the sum of all future next-hop durations should align with the total remaining EDT. We formalize this by defining an aggregated target d_t^{agg} as the scaled sum of future ground-truth durations Δh_j :

$$d_t^{\text{agg}} = \frac{1}{C} \sum_{j=t}^{T-1} (\Delta h_j - b) + b \quad (5)$$

where C is a constant for unit conversion, and b is a bias term. We then impose a consistency loss, $\mathcal{L}_{\text{cons}}$, that penalizes the L2 distance between the model's expected EDT prediction $\hat{d}_t$ and this self-supervised target d_t^{agg} :

$$\mathcal{L}_{\text{cons}} = \frac{1}{|\mathcal{M}|} \sum_{t \in \mathcal{M}} \|\hat{d}_t - d_t^{\text{agg}}\|_2^2 \quad (6)$$

This forces the model to learn a more fundamental and self-consistent representation of time.

2.2.3 Dynamic Loss Scaling. To adaptively balance the magnitudes of the different loss terms, we employ a dynamic weighting mechanism. For each loss component $\mathcal{L}_k$ (where $k \in \{\text{cons}, \text{reg}\}$), we maintain its exponential moving average, denoted as $\bar{\mathcal{L}}_k$. After each

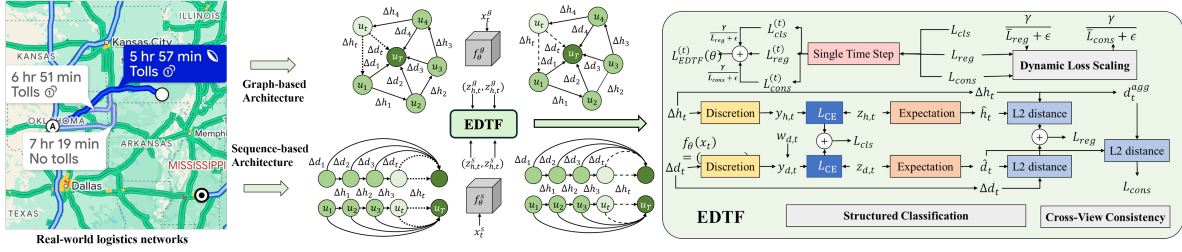


Figure 1: An overview of the EDTF framework. f_{θ}^s and f_{θ}^g denote different architecture approaches

training step, this value is updated based on the loss computed over the current mini-batch $\mathcal{B}$:

$$\bar{\mathcal{L}}_k \leftarrow \alpha \cdot \bar{\mathcal{L}}_k + (1 - \alpha) \cdot \left(\frac{1}{|\mathcal{B}|} \sum_{u_t \in \mathcal{B}} \mathcal{L}_k \right) \quad (7)$$

where α is the momentum coefficient. The dynamic weight for this component is then defined as $\frac{\gamma}{\bar{\mathcal{L}}_k + \epsilon}$, where γ is a scaling hyperparameter and ϵ is a stability term.

2.3 The Complete EDTF Formulation

Integrating all components, the EDTF framework is guided by a unified, adaptively weighted objective. The complete loss function, $\mathcal{L}_{\text{EDTF}}$, is the average loss over all valid time steps $\mathcal{M}$ in a sequence:

$$\mathcal{L}_{\text{EDTF}}(\theta) = \frac{1}{|\mathcal{M}|} \sum_{t \in \mathcal{M}} \left[\mathcal{L}_{\text{cls}}^{(t)} + \lambda_{\text{cons}} \cdot \left(\frac{\gamma}{\bar{\mathcal{L}}_{\text{cons}} + \epsilon} \right) \cdot \mathcal{L}_{\text{cons}}^{(t)} + \lambda_{\text{reg}} \cdot \left(\frac{\gamma}{\bar{\mathcal{L}}_{\text{reg}} + \epsilon} \right) \cdot \mathcal{L}_{\text{reg}}^{(t)} \right] \quad (8)$$

where the superscript (t) denotes a loss term computed at a single time step t . The terms $\mathcal{L}_{\text{cls}}^{(t)}$, $\mathcal{L}_{\text{cons}}^{(t)}$, and $\mathcal{L}_{\text{reg}}^{(t)}$ correspond to the classification, consistency, and regression losses defined in §2.2.1 and §2.2.2. Finally, λ_k represents the base weight for each component.

3 Experiment

3.1 Experimental Setup

Dataset.¹ We construct a large-scale dataset from the real online shop platform, containing over 10 million logistics trajectories from Country A. To ensure a realistic evaluation that accounts for temporal effects, we partition the data chronologically. The training set consists of 22 weeks of data (Jul.–Dec. 2024), followed by a 2-week validation set (late Jan.–early Feb. 2025) and a 1-week test set (mid-Feb. 2025). The final dataset comprises over 200 million EDT events. Key statistics include an average next-hop duration (Δh_t) of 36.79 hours, an average EDT (Δd_t) of 4.99 days, and an average sequence length (T) of 19.77. **Preprocessing and Metrics.** We first remove duplicate events. To handle outliers, we then apply two distinct clipping strategies based on the 95th percentile. For ultra-long sequences, we truncate their length T by keeping only the first 35 events. For extreme durations and delivery time, we cap the values using the transformations $\Delta h'_t = \min(\Delta h_t, 80 \text{ hours})$ and $\Delta d'_t = \min(\Delta d_t, 40 \text{ days})$. We evaluate performance using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE).

¹Due to company policy, specific details cannot be disclosed at this time.

Implementation Details. Models are trained using the Adam optimizer with an early stopping patience of 5 epochs. Unless otherwise specified, the hyperparameters for our EDTF framework are set as follows: $\lambda_{\text{reg}} = 1.0$, $\lambda_{\text{cons}} = 0.2$, $\beta = 10.0$, $w_{d,1} = 5.0$, $\alpha = 0.9$, $\gamma = 1.0$, and $b = 3.0$.

3.2 General Effectiveness of Plug-in EDTF

We evaluate EDTF's effectiveness as a plug-and-play framework. We replace the native loss functions of five diverse baseline models with EDTF and present the results in Table 1. The data shows that EDTF consistently yields significant performance gains across all architectures, confirming the viability of EDTF. The improvements are particularly pronounced for sequence-based models. For instance, EDTF reduces the RMSE for DeepTTE by 2.42%. Most notably, when applied to TTSOL-V1 (the production model at the real online shop), it achieves a remarkable 8.28% reduction in RMSE. On a web-scale dataset, performance gains of this magnitude, validated by tight confidence intervals, are of substantial business value and provide strong evidence for the universality of our framework.

3.3 Ablation Study and Comparative Analysis

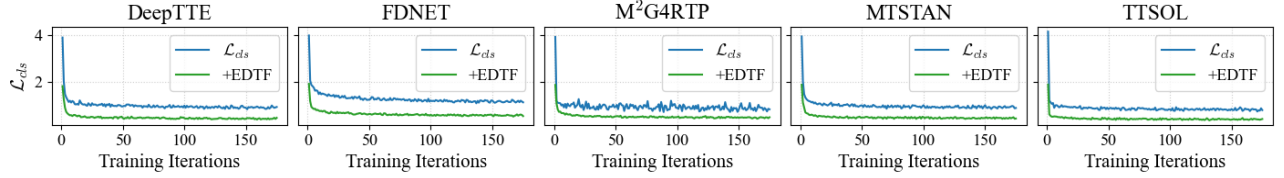
To evaluate the contribution of each component, we conduct an ablation study on the TTSOL-V1 backbone (see Table 2). First, the results confirm that every component is essential: transitioning from a standard regression baseline to our classification approach yields significant performance gains, while incorporating ordinal and consistency losses further optimizes metrics to achieve the full EDTF model's peak performance. Second, EDTF outperforms related SOTA methods in accuracy; the tight confidence intervals in Table 2 confirm the statistical significance of these gains. Specifically, EDTF achieves lower MAE and RMSE than specialized ordinal (Coral [5]) and multi-task alignment (PCGrad [7], OT [8]) techniques. Third, EDTF is substantially more efficient for industrial deployment, with the "Time" column indicating significantly lower training time per epoch than all compared methods. This combination of superior accuracy and high efficiency directly supports agile, low-cost model iteration in dynamic web environments.

3.4 Convergence Acceleration

We also validate our claim that EDTF's auxiliary losses provide richer gradients to accelerate training. We compare the convergence of the primary $\mathcal{L}_{\text{cls}}$ on all five backbones within a single training epoch, using a batch size of 512. Fig. 2 clearly demonstrates that for every model, EDTF leads to a faster initial drop in loss and converges to a significantly lower final value. This provides strong

Table 1: Performance comparison of baseline models with and without the EDTF framework. "w/o EDTF" denotes the model trained with its original loss. "Improve" indicates the percentage reduction in the metric. CI is the 95% confidence interval.

Model	Note	Framework	Params	MAE ↓	CI	Improv.	RMSE ↓	CI	Improv.
DeepTTE [6]	A spatiotemporal network for urban vehicle EDT.	w/o EDTF	2.24B	0.661	1.07e-3	—	1.811	5.45e-3	—
		+ EDTF	2.45B	0.641	1.05e-3	3.03%	1.768	5.46e-3	2.42%
FDNET [2]	An attention-based model for EDT in end-stage food delivery.	w/o EDTF	0.84B	0.835	1.11e-3	—	1.937	5.25e-3	—
		+ EDTF	0.97B	0.820	1.09e-3	1.77%	1.895	5.27e-3	2.15%
M ² G4RTP [1]	A graph-based model for inter-station parcel EDT.	w/o EDTF	1.00B	0.667	1.08e-3	—	1.828	5.42e-3	—
		+ EDTF	1.05B	0.657	1.06e-3	1.51%	1.798	5.43e-3	1.66%
MTSTAN [10]	An attention-based model for vehicle route EDT.	w/o EDTF	3.05B	0.724	1.09e-3	—	1.867	5.34e-3	—
		+ EDTF	3.45B	0.721	1.08e-3	0.35%	1.850	5.32e-3	0.93%
TTSOL-V1	An attention-based model for parcel EDT in a e-commerce platform.	w/o EDTF	1.99B	0.641	1.13e-3	—	1.885	5.99e-3	—
TTSOL-V2		+ EDTF	2.20B	0.610	1.04e-3	4.93%	1.729	5.49e-3	8.28%

**Figure 2: Convergence of the classification loss ($\mathcal{L}_{cls}$) within the first epoch. "+EDTF" denotes the model trained with EDTF.****Table 2: Ablation and comparative analysis on the TTSOL-V1 backbone. Time is reported as hours per epoch (with mixed-precision training). CI is the 95% confidence interval.**

Method	Time	MAE ↓	CI	RMSE ↓	CI
<i>Ablation Study:</i>					
(1) MSE	1.96h	2.574	2.09e-3	4.177	5.40e-3
(2) $\mathcal{L}_{cls}$	2.38h	0.641	1.13e-3	1.885	5.99e-3
(3) $\mathcal{L}_{cls} + \mathcal{L}_{reg}$	2.39h	0.636	1.07e-3	1.794	5.47e-3
(4) $\mathcal{L}_{cons} + \mathcal{L}_{cls}$	2.39h	0.639	1.07e-3	1.802	5.47e-3
(5) EDTF (Full)	2.41h	0.610	1.04e-3	1.729	5.49e-3
<i>Comparison:</i>					
(6) Coral [5]	3.50h	0.666	1.03e-3	1.750	5.43e-3
(7) PCGrad [7]	7.34h	0.620	1.01e-3	1.708	5.47e-3
(8) OT [8]	12.15h	0.631	1.06e-3	1.787	5.48e-3

empirical evidence that EDTF enhances training efficiency, a key requirement for agile deployment in industrial web applications.

3.5 Real-world Impact

The ultimate validation of our framework is its deployment in a live production environment¹. An EDTF-enhanced model, **TTSOL-V2**, demonstrated significant performance gains over its predecessor in online tests on the real online shop platform. Consequently, the new model has been fully deployed, validating the industrial feasibility and practical value of EDTF.

4 Conclusion

We introduced EDTF, a plug-and-play framework that improves EDT performance and deployment efficiency via a reformulated learning objective. Extensive offline and online results confirm its

ability to achieve faster convergence and higher accuracy. Our approach serves as a scalable paradigm for various industrial web service applications.

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